

ActInf Textbook Group ~ Cohort 7 ~ Session 22 (Chapter 9) ~ 4/3/2025

TextbookGroup/ParrPezzuloFriston2022/Cohort_7 | Cohort 7 ~ Session 22 (Chapter 9) ~ 4/3/2025 | 56:17

Yeah,

[Music]

okay. It's recording.

Um, so for for Joseph, for Daniel, for the rest of the institute who's uh following along remotely uh or via YouTube, um welcome to the Active Inference Institute. Uh we do a uh weekly textbook group readings uh twice on Thursdays, once in the morning, once in the evening uh if you're if you're roughly in the Western Hemisphere. And um yeah, so so today we're going over chapter 7 in cohort 7. Um and we'll be talking about essentially the exemplar discrete state space models and active inference. That would be hidden markoff models and uh partially observable markoff decision-m processes. We have a nice T-ise example which is pretty classic in in the field as well as in kind of behavioral and and and and neuroscience and neur neuroimaging studies in general. And uh again we're in the second half or part two of the book which is highly dedicated to to constructing uh essentially active inference models as opposed to part one which is much more focused on theory and and definitions. Um, and so it's for anyone who's interested in computational modeling, whether it be doing it themselves or getting a clearer grasp of kind of what are these computational models or cognitive models that we're

putting together in order to say collaborate with a uh team of of people who do do cognitive modeling, then it it's very worthwhile to spend time with the second half of the book. And then furthermore, there are just there are many discrete state spaces in in a in a very general sense, in a in a physical sense, in a kind of decision- making and in in and human belief sense. It's referring to scenarios where essentially if we think about it in data, it's sort of like categories. It's thinking about binaries. uh it's not so much thinking about uh fully continuous number series such as a range over one through a million where we could have so many different uh different values including with the decimal points uh potentially an infinite number of values instead we're looking at say RGB color scales on old television so red blue green um we could have a distribution between the three of them in the sense of how much red how much blue and how much screen. Uh but nonetheless, it's it's a kind of um discrete uh categorization process. So, um yeah, normally I give a rather long uh kind of chapter summary, but I'm happy to just kind of open up the floor if uh Joseph, you have any particular questions or things you'd like to discuss here.

Um I I would like to hear your summary. Um I also do have some questions. Um I haven't fully been able to understand

um for example figure 72. So it if we could like I I I guess I just don't understand what's plotted and and where the lines are coming from. But um yeah, I I'm I'm very interested in in trying to build these models and understand how to build them and um and so this seems like a really critical chapter for for me.

Yeah. Yeah. No, that's great. Um yeah, figure 72, which I I personally have not spent a lot of time with recently, so coming back to it is rather interesting. Yeah. Yeah. So the the this is where the chapter it essentially opens with hidden Markov models uh before we jump up to uh partially observable Markov decision-m processes.

So it's like we have um I believe my is my screen still being shared appropriately or can you not see anything? Uh no I don't see your screen.

Okay, got it. Thanks for that. Just realized I doing a little bit of uh display gymnastics. Um that should have done it. So this is and a quick reference. This is the KOD for the textbook group. And so we have this available to all members and we have u not just a overview of the textbook group and you know dedicated uh meetings and and time availability and and all those sorts of things for the separate cohorts. We also have a pretty good breakdown of the active inference textbook especially since it became open source.

So, um, so yeah, figure 71, I know it's

not the one you referenced, but just to make sure we're like familiar with it enough, this is essentially a hidden Markov model. And so understanding kind of the core dynamics here, uh, we ultimately only have three different components in this model. So we have D , our prior over initial states. It's kind of like what are our beliefs about states whenever we come into a situation as it were. Um and then from there uh that gets linked to uh observations that we receive right and the relationship between what we believe and sort of what we see is linked through this this likelihood matrix A . So it's represented as kind of this broad categorical probability distribution that is conditional uh observations given our beliefs S or in states S and then there's also a temporal dynamic over time right so we have the B matrix which is kind of our beliefs about how transitions change from one to another so that's why we have this sort of state at $t - 1$ to state at t to state $t + 1$. So it's this this temporal dynamic playing out over discrete time steps. Um and so from there you can move to

7.2. A little bit of context. So this is this is an example that relates to the idea I believe it's of of someone who's playing uh an instrument and they're trying to actually play a song in accord with the the script, the sheet music that they're reading. And so we can quickly imagine a scenario where there's

someone who is still learning how to play. This is practice for them. Even if it's someone who's very skilled, they're maybe learning a new piece for like, I don't know, an orchestra performance they'll be doing soon. Um, and so there's all kinds of room for error, right? And to accidentally hit the wrong notes at the wrong time. Maybe you're still on beat, but uh but you know, you hit an A rather than an A flat. And so that's sort of what this example is about is being able to use the dynamics of a hidden markoff model, including what notes play over time and then also what does the musician believe the note should be versus what we actually hear. Um, right. And so by being able to like hear and it's obviously a very kind of abstracted example. There's a lot more to it there. We're not saying anything about what the sheet music looks like, but here we're just simplifying it to saying the musician has a belief about what the note should be versus what they actually observe that they're they're playing. And

so with this figure 7.2, I think the the quickest way of kind of getting at what the authors of the textbook are trying to demonstrate for us as sort of an illustrative example is this upper right. This is like beliefs about hidden states. So the agent believes that the first note uh in terms of over time should in fact be the first note as in it's in row one. So that this this row

is like its own set of notes. Row two is the uh yet another note. So it could be read as like the top row is A. The second row is A sharp. The next row is B. The next row is C. Not to confuse anyone who's not familiar with scales, but we there is no B sharp essentially unless you're concerned with naturals and all that. So, um, so each row is its own note. And then I here it's kind of like the the musician believes based on the sheet music that they should start with the first note A and then move to A sharp, move to B, move to C, and then go back to A for the fifth note. So, they're just kind of descending a chromatic scale before they get back to A again. That's what they believe should happen. Meanwhile, with our O's or observations, what really happens is that we get an A, we get an A sharp, and then rather than getting the expected B, which is here, we the the actual observation is that they realize, oh no, they hit the same note twice in a row, which is not correct. But by the fourth time step, they get back to what the note should be. And then here uh at time step five they also hit the correct note. So kind of what's what's happening here is that there is at at S3 we would say that this is prediction error, right? And so with whenever the agent infers their posteriors over hidden states here they're so conditioned to believe that it should be the right note that that they kind of shaded this in as if it's correct. But there's this

grayness here. It's very it's so light
but but essentially the idea is you know
this is the note that should have
happened their prior their observation
however gives them evidence that it's
actually this gray space this is maybe
actually the note that they heard um or
that they believe really happened. So
there there are aspects of this that can
seem kind of counterintuitive, but
really just being able to grasp that the
upper right are their beliefs about
states. The lower uh the lower right is
the observations they actually receive.
And then a good reminder is you know in
a discrete state space setting it's like
oh we're not looking at
you know the interim between A and A you
know some kind of microonal situation.
we're just looking at uh we're assuming
there's a and then there's a sharp and
then there's B and then there's C and
ultimately they're that's the western
scale of 12 notes um in the way that I'm
describing the example. So now these
these graphs on the left are a little
more complex um and it will be good to
spend a little bit of time with them.
I'll first just like read what it says
in the textbook. So, so we'll go with
lower left. We're looking at negative
free energy gradients or prediction
errors. The rate of change of the
beliefs in the upper left
plot is determined by the value of these
errors at each time step. And so what
we're seeing these are negative free
energy gradients. In terms of active

inference, we're wanting to minimize prediction error. We're wanting to minimize variational free energy. So if these are actually negative gradients, then that means that at each time step, we should actually want these to be rather large um if they're negative. And so the the smaller they are, you know, it's kind of like saying that it's sort of the the worerer scenario. And then that seems to kind of constrain the agent's ability to to be confident or or or kind of understand the the situation. So um see and then upper left beliefs posterior probabilities about each note in the sequence at each time step. Yeah. So the these are the posterior probabilities. It's kind of like their ability to understand a particular note here of what it should have been h has gone down. Right? So there's a kind of constraint there that's been applied. And then once they start to hear the correct notes again, their confidence kind of it kind of increases. I will be honest, these kinds of these two graphs on the left, they're much more rarely used today in active inference like these specific types. It might be because of this sort of kind of kind of uh counterintuitive nature because more often than not, we actually are looking at you the inverse of the situation. So we're looking at positive free energy. And so actually this would look quite bad. it would look like step three is actually, you know, a rather good thing

in terms of minimizing uncertainty and it's the other way around. But I I think that the authors who who compose this are really trying to get at a particular kind of point by representing it this way. Um I I think I'm still maybe a little lost about so um you said that just focusing on this top left um the the y axis you said is probability. So okay so I I get that and what are all the different lines? Um are are these trials or um one of the things yeah one of the things I'm really wanting to understand better is um uh how the agent actually is learning and and maybe that comes later on. This is after this example. But um where do do we adjust the weights of this of the the in these matrices um according to the result we get from our minimization step? Oh yeah. No, that that's an excellent question. Very relevant. So so in this example, they don't actually apply learning at all. So this is essentially like a static model. Yeah. Um, and I think they're just using it to try and kind of get at the more inference level dynamics as opposed to learning. And so I I know that this is, you know, this is pretty familiar terminology for folks who do other forms of kind of computational modeling, data science, deep learning. Uh, I I heard you say that, you know, referring to it

as adjusting the weights. All of that is very relevant here. But so, so for this particular model, yeah, there is no learning. What would what could happen is that refer to this again. Um this what typically what an agent learns are these kind of probability distributions over time. And so D you know your your prior beliefs in the Bayesian sense those could be updated. So you could the agent could effectively learn its D matrix. um A could be learned, B could be learned actually any of its parameters, including C and E, which C is just a prior over observations. E is a prior over over policies or actions. Um so, so that way you know that's what's missing from this and it's also missing from this because we have no policy inference, right? So the HMM is kind of like can be viewed as a truncated version of the POMDP which is exactly equivalent until we get to here this connection to to PI or representing policies or action. And so so an HMM is kind of like perception and no action. uh partially observable markup decision-making process is perception and here we have the connection to actions that it can take. So we get these these kind of relationships where suddenly state transitions the B matrix recall earlier 7.2 two the B is simply what is the next state going to be given the current state. Whereas here we now

have yeah we now have what is the next state as well as an action I take. So that is sort of the kind of the the agency so to speak that this this agent has um in its ability to kind of infer which actions to take and furthermore um e the prior over policies I mean that so that can be learned that gets incorporated. C preferences prior over observations there those are more often left static. So those are more often left static in the sense that they become a kind of attracting point for particular observations.

uh in that case that kind of supplies the agent with a sort of goal. Um it kind of like it's seeking out particular observations that it it it prefers or expects. Um in the sort of theoretical literature, the idea is that it relates to the notion of homeostasis. So, I prefer to observe my blood oxygen level being at a certain point or I prefer to not observe my stomach growling because usually it's a sign that I'm hungry and I need to do something about that which in a that context it would that that C would then inform what I should do via the B matrix of oh my stomach growling must mean I'm hungry and I know that if I eat food I will s I will no longer be hungry at the next time step, so to speak. Um, so, so again, any of these can be learned, and I know that there's a nice figure that kind of hits at the point of this, but I don't want to get us too lost in the weeds.

Um, hybrid

models, the essentially what happens is that we end up with this kind of hyperparameterization of all of those model components. Okay, I'll stop I'll stop scrolling around because there might be that we missed that one. Still relatively new. Um, so you would have like a prior on D which itself is in the discrete state space setting. So it would be it would just be like a kind of so it' be a deerishlay distribution. So so it' be like a categorical distribution but the values can in fact exceed one. they don't have to sum to one. You could almost think of it as a kind of like dearishly count system where it's like oh I I you know I have my D matrix but then I've also experienced like more of this state in many more time steps. So it it implies that there you that an agent actually has a history of states and a history of observations that it's kind of accumulated and then that acts as a kind of hyperparameter on these matrices which then proceeds to modulate their values and then learning them itself will involve free energy minimization. I it's just applying another free energy minimization rule. Um and so and so like minimizing free energy is sort of the global uh functional or optimization problem here for just about everything like state like inferring states occurs by minimizing variational free energy inferring policies heavily relies on uh minimizing expected free energy which uh earlier on

in the textbook I think it's somewhere
in the realm of chapters 2 through or uh
there's kind of a broad
breakdown about those terms. So we
should be able to find those. We have
the equations as well. Yeah.

Great. So something we'll notice very
quickly between F and G. F is going to
be for again state inference which ends
up being treated as a separate process
related. But if if you were to code this
like if you were to write this in code,
you would actually write like separate
logics and even executions of state
inference versus policy inference. State
inference will involve min minimizing f
which without getting too lost in the
various ways we can break this down. We
could look at it as complexity minus
accuracy which is the sort of classic
you know machine learning imperative of
how do you create a model that can both
be generalizable and not excessively
complex and overfit um but we need it to
be accurate. So we don't want it to
heavily underfit either. Um so the the
complexity term there is just what is
the kullback divergence between your
posteriors what you think now versus
your priors what you thought before and
then you want to also keep accuracy
high. So we want to be able to find the
expected value of the log of essentially
observations given x. So, so all this is
to say each one of these terms depending
on what the sign is in front of them um
we'll we'll want to either maximize or
minimize. So we'll want to maximize

accuracy right because it has a negative sign in front of it. Vice versa for complexity we actually want to minimize that because at the end of the day we come back to f we're just trying to minimize that.

Um similarly for G. But the interesting thing with G is that whenever we look at just the top

line information gain and pragmatic value we notice

that there are negative signs in front of both of them and we're trying to minimize G. So we actually here want to maximize information gain. Meaning the the the expected value of the kullback divergence between sort of what you your posteriors what you what you think now conditioned on what you've seen and what you've done and the the the divergence between that and just what you think now versus or conditioned on what you have done. So this is kind of like saying oh now that I have actually observed the outcomes of what I have done and what I believe now versus simply just relying on you know sort of what I've done. It it kind of prioritizes this idea of, oh, I need to collect more observations, and I actually want those observations to be quite different from what I you I'm used to seeing before, what I'm so used to that I can practically sort of marginalize them out. Right? So that that's sort of the exploratory behavior we see in active inference. And furthermore, just the fact that we have this functional that actually allows for

both exploratory behavior and pragmatic value, which is simply your observations conditioned on C. So like I mentioned earlier, C it's often used for defining like goals that the agent has. It's kind of like the what you expect is what you want, which is what will help you maintain homeostasis or it's what will help you achieve that goal that you have in the far future or however you want to look at it um or goal in the immediate future. And so you want your observations to essentially like you you want what you see to actually be accurate versus like your priors over observations, right? So that that's that's kind of I don't know if you're familiar with reinforcement learning, but anything like epsilon greedy behavior or having to like define manual rules such as like oh you should always exploit with a probability of 0.9 and you know 10% of the time just do something random and see see what happens. instead instead this is what defines exploratory behavior is directly implicated in computing minimizing expected free energy. So the that's the claim in the textbook that these sort of naturally balance out which is not a sentence that really should be said that way in the sense of you could still have a scenario where a policy is evaluated such that it might not have any information gain at all. It's just the fact that both are computed and so there there is no manual definition here.

We're just relying upon minimizing G and these particular equations. I might have gone much more deeply into that. And I I hope I didn't miss No, I I really appreciated that. I liked your explanation about um the the the meaning of those different distributions that you're measuring the Kullback divergence. Um that that was very helpful actually uh especially when you were talking about the uh information gain and the fact that you know one is conditioned on why hat or why tilda and one is not. Um I I had I just sort of not connected those dots before. So um yeah.

Yeah. No problem at all. Can you say again, I think you said it already, but can you say again the difference between P and Q ? Is this related to the world model versus the uh agents model? Like the truth, the true world versus the agents what's in the agent's head or is it something else? Yeah, sure. No, good question. So, um, if we were to kind of use the phrase, is it just in the agent's head or the world, it's all of this would be in the agent's head. Okay. So, yeah. Yeah. So, and then uh in the sense that like this is the agent's beliefs about states. This is the these are the this is kind of the breakdown of the overall rule of computing free energy over states. So, states as in my my beliefs about the world and then G will be used for inferring what policy should I use. So, that's as opposed to

state inference, it's policy inference,
right? Okay. So the and and and all of
it is in the in the agent's head as it
were. And then for Q and P , so Q is
always used they they always use that to
denote posteriors. So this is sort of
like what what do I think after a given
instance versus you know and versus what
did I think before, right? So there's
this kind of model inversion process
that that that happens where where we
compute f by sort of fig sorry sorry
about

this figuring out our posteriors
figuring out our relative to our priors
and that's what allows us to kind of
score the final like what is our final
belief now that that we have all of that
and we have now minimized the free
energy right so just basically employing
basian inference but in the context of a
fuller functional equation.

Yeah. Yeah. So, so you know there are a
lot of interesting ways that the these
terms get broken down and I know that
like for for a lot of folks who get
involved with the institute and their
learners uh especially those who are not
terribly uh you know comfortable with
math or or probability theory it's sort
of like these things can be very
intimidating right to just like quickly
see like you know let us tell you all at
once all these different ways you can
break down G whenever you're still
trying to learn what this little C is
over here and how what that means
relative to anything else, right? And

what's the tilda about, right? So, and I'll also I guess I'll mention that the tilda is uh so G what you're doing here with G and the the whole G over π is you're you're inferring your beliefs about each policy you have available over some inference horizon which is why we call it expected free energy. It's that it it also is variational in the sense of like there we're relying on different kinds of you know approximate methods variational methods here but the the the expected part is emphasized because if if I notice I'm hungry and I'm trying to decide what to do it's like well if I need to if I decide one policy I have is that I need food there's going to be a series of steps that I'm going to need to take. Do I have food in my fridge? Is it at the store? If it's at, you know, if I decide going to the store is what I want to do, then like what are the intermediary steps of what I need to do from between now and then. Um, surely I don't think about every single thing like the entire process of what it will be like for me to walk down my apartment complex's hallway, but more discreetly I know that I will need to say stand up and then I'll need to leave my apartment and then I'll need to travel or take transit or otherwise to the store and so on. So, so it's kind of like the tilda is just acknowledging that this is a sequence. So, what what are the kind of expectations I have over what will happen if I follow a particular policy

and and how I think it will occur. So um
which is why we also don't see the tilda
up here because this is more about
momentto- moment inference which again
relates back to earlier the point about
learning right because it that's that's
an important distinction between the two
and active inference is uh state
inference is just about you know in real
time you know you think about again the
the musician playing notes in real time
and and it's very directly related to
their ability they hear the note that
they're playing versus what they think
it is. Um, so that's that's one way of
kind of just more conceptually
conceiving of uh of these things.

So, okay. And and so I guess um
especially with G, it seems to me that
the reason you would choose one
breakdown over the other, maybe with not
just G, but all of them, but um is
because you're you're trying to figure
out what's actually computable, like
based on the computing resources you
have.

Maybe it's a lot easier to compute um
that fourth line there with the expected
energy and entropy than the other
options. And in terms of uh what's
actually solvable
um that that's possible. I mean it's
never made very explicit why we have all
of these different breakdowns. The the
crucial thing is that like the way the
math plays out, it's necessarily the
case that this first line like one
version of G is information gain minus

pragmatic uh negative information gain
minus pragmatic values G . It's
necessarily going to be equivalent to
the value here um given the the
equation, right? And so it's sort of
like now another sort of tricky thing is
that the this is sort of they will be
less than or equal to these other terms.
So th this allows for almost I don't
want to get too abstract and mix things
together, but it's almost like this is
an additional like this is the the the
lower uh excuse me the upper bound of
the upper bound on like expected future
surprisal right it's because we already
know that um f from definitions of
surprise like f is just an upper bound
on surprise that that works out
mathematically and so it's kind kind of
like saying well these are the upper
bounds on surprise expected in the
future and these are kind of the upper
bound to the upper bound because of this
sneaky uh less than or equal to sign
here right um so so another thing is
that it's sort of like the another
aspect of this beyond that is u the
authors try to supply their own
definitions of the different ways of
breaking these down. Um, and so expected
free energy in just about
every as a broad generalization, but I
would I genuinely think maybe 80% to 90%
of all literature you read in active
inference, most people are just going to
focus on describing it this way,
information gain, pragmatic value
because it comes off as the most

intuitive and it's also the most interesting in terms of viewing things uh in like explore exploit paradigms. It's just this is the clearest way of kind of um kind of expressing that and so it's it's just intuitive like oh yes I want more information that actually increases like what what I thought before versus what I know now. It's almost like I'm seeking out. I hesitate to use the word learning because of its its technical connotation here, but sort of like I want to learn more or I want to find more. You know, it's a sort of a someone who wants to have fun doing something new for the sake of it. You know, I'd sort of think about that as a more colloquial way. And then pragmatic values just, you know, is what I'm seeing or what I expect to see is it going to be aligned with my priors. So, so is what I is this particular action, this particular policy I'm evaluating, like going to the store to get food, is that going to help me achieve observations that match what I want, which is attaining food and kind of watching in myself eat it and then watching my stomach no longer growl is a kind of odd way of saying it, but it it's closer to the the ideal of of how it would

Yeah. Um I I think part of the the break the breakdowns here is just sort of it's taking into account so like with expected ambiguity and these other terms uh some of them additionally get introduced in in chapter 7 as well. Um

posterior predictive entropy those kinds of terms. is kind of like just saying like oh it's it's sort of like this this idea that you know there's a lot of other dynamics of how we could view these plane now. So expect expected ambiguity is like um oh that that for example is sort of what is the entropy of what I expect to see versus what I expect will be going on. So um you know if if I expect that the it will be light outside because I believe it's daytime and I expect to see that it's dark outside. So I believe that it is uh nighttime then it's sort of like we have this nice one to one discrete mapping. It's light out it must be day. It's uh dark out it must be night. Um, however, if uh if we if we expect more of a one to many mapping such as uh oh, it's light outside, it could be day or there or excuse me, it's dark outside, it could be night, or it could be a solar eclipse right now, right? And I I'm just not sure which. I didn't pay attention to the news. I don't know like what's going on, right? So um that's suddenly where you get more of a one to many mapping between like what could this particular observation mean? Um and here it's a it's a positive value. So we'd actually want to like minimize that kind of that entropy of that situation. Thus the expected ambiguity. Um and then risk over outcomes. So these risk terms that get related to because there are a lot of

people who go into this field not just because they're interested in say neuroscience but they might be interested in other fields. I I myself have a background in um I did my uh masters at at Chicago and this very broad social sciences program. That's part of what part of what brought me here uh including an interest in economics. still looking at risk. And so this is like your posteriors versus your priors. Like what what do you believe right now about what you would expect to see versus versus pi versus like what what do you believe after? Right? So what what do I expect to kind of see from what I do versus what I I I'm used to sort of um and finding the kale divergence between those, right? and you want to minimize that, which is which is a little different. So, it's just there's so many ways to break it down and so many different ways to attempt to put into words why these equations work the way they do. That's sort of the the trick, isn't it, of not just active inference, but sort of cognitive and computational modeling in general. like well if we have real data and we have real approximate methods of mapping things like beliefs and actions and inference learning processes what how do we put into natural language what any of those things mean.

So, it's a I think it's a fair attempt and I I personally like many other people kind of prefer this top one, but

it is interesting breaking the rest down. And then it's it's kind of like the further you go down the line here, we get closer and closer to theoretical physics, which is yet another kind of partner field of academ's um research, right? So just having energy terms that yeah my my background is in physics and I think that's a large part of why this um appealed to me is a I'm I'm working on a robotics company. So um that's how I came into this and found it and and why I think it attracted me was the analogy to physics. And um I'm hoping to figure out some ways to reduce the amount of data that it requires to uh for an agent to learn a task or to decide to do a a sequence of motions or something.

Yeah. No, absolutely. I mean, it's uh I don't know if you skipped ahead to um chapter 10 in this textbook at any point, but I think they spend some time on on robotics and and you know the the more theoretical aspects of like cybernetic processes and sort of feedback loop operations, but then like suddenly with active inference and this very broad interdisciplinary aspect, it's sort of like because it can go so many ways it because it comes out of neuroscience. I mean one way one kind of funny takeaway and maybe it's not funny but I I I think it's kind of fun is uh like a lot of this relates to the brain and inference processes as well as looking at behavior

like the T-maze example that we get in this textbook. So uh you know like a similarly to actually having a real mouse in a lab but then if computational psychiatry plays into this a lot and it's actually like it's a growing field and many people are trying to cognitively model and I'm working with a research group here as well uh but we're working on a on creating an active inference model of PTSD actually and uh sort of what are the behavioral dynamics there and and sort of the belief dynamics. And can you find a a generalizable but still approximate um sort of model of these these dynamics in relation to neurobiology and then in future like being able to kind of fit that model to real empirical data that could be collected from you know willing participants uh who have PTSD. Um, and and all this is to say, the reason why I'm kind of going on about it, despite the fact that you we started with robotics, is that it kind of in an interesting way, it's almost like like is a bad machine learning model or reinforcement learning model like if it's bad, should we view that as a kind of like like a pathology? Like can we view suboptimal behavior as sort of something like pathological behavior? So I I just think that's a that's a fascinating kind of take and I I've talked to a few people who are more focused on sort of sort of optimizing models rather than you know they're not

worried about attending to human beings and modeling that they're wanting to create very efficient you know kind of engineering based you know tools and and and so on right and we've always had interesting conversations about that because because in active inference it's like there's a term that's used sometimes.

Um the term is a hyper precise prior. And so that that would kind of be like a scenario

where

to bring it back a little bit. It it could be for example a situation where skip it because you see figures.

So like imagine an agent, you know, imagine our musician example. Again, we don't have even have to worry about policies just yet. Um, imagine this agent like had a very precise hyperparameter like on their A, like a very strong prior for their A matrix and for some reason they're just learning it at either an incredibly slow rate or not at all. Um, and say the A matrix was really biased. So like with the with the this example like say the agent was heavily biased to assuming that like if their like their these beliefs are necessarily like onetwoone with with these to where even this kind of light gray space didn't appear so to speak.

It's like they wouldn't track the prediction error because they believe in the basian sense so strongly that this is in fact how things work, you know, perceptually given that the HMM is just

like a perceptual model. Then it's kind of like saying like oh that's you know it it in in robotics that I mean that would be awful. That means it's not able to like read the data streams in a way that allows it to make any kind of like flexible adjustable inferences. instead it's it's going to just assume whatever it sees regardless of what it is just reconfirms like its beliefs about you know what's going on and if this were a POMPP then also what it should do in response to the data stream it's receiving but uh simultaneously the pathology bit is sort of like oh that would for a musician that would mean that they believe they're really good at music all the time right because they're they're sorely convinced that their beliefs about what should be heard or what should be played or what's going on uh necessarily match the incoming data stream. And so the incoming data stream will just kind of like it won't the variations in it won't really matter. It's like there will still be won't even hear that they made a mistake essentially. They'll they'll believe that it's right.

Exactly. And so, not to, you know, not to sound, you know, insensitive or or or phrase this wrongly, but it's, I mean, if you start thinking about things like schizophrenia or psychosis, where something in reality is not going on at all, but the person believes it is, it's like in a certain way, it's they're they're perceiving reality in a way

that's heavily biased by these kind of priors that they're coming to the situation with, right? So you're sort of stuck in that uh if so long as the your your inference process is kind of you know kind of path pathological in that way. So

um so yeah it's I mean with robotics I'm not that's not my my key field as it were. So I can't say too much other than anytime I've skimmed it, there's always something being something interested being said and that you know with a lot of these models and various areas um they're highly performative including in when comparing them let's say to reinforcement learning because it's another way of looking at active inference especially especially POMDPs or any any model that is actually augmented with the capacity for action.

We're effectively just looking at okay here's an agent who receives observations and infers an action it should take. Um if we just look at it as a markov decision making process like remove the partially observable part then it would just kind of be like they are it's as if they already know what the state is. They observe a state. It's sort of more onetwoone. we don't have this

extra observation versus states part, right? But uh but but nonetheless, I mean, this is still the canonical bit because I mean this is as far as kind of the neuroscience bits go. I mean, we're always in a partially observable

environment, right? So we we don't always know what an observation means and we haven't even said anything about like you know the possibility that the sensory receptor itself is bad whether it be you know a person who's nearsighted and so anything far away is just very hard to see or it be um you know a robot that has a a receptor that is just like not functioning correctly or um an EEG headset that like you know someone left the ground uh you know they didn't set that up right and so it it it sees really high amplitudes not because of the person's neural activity it's because it's getting like supercharged um so in a inaccurate way so that that's sort of the part partially observable bit um they do I mean anytime I look at model comparisons it's just you know people do very regularly benchmark Um yeah, they they regularly benchmark these models against reinforcement learning models. um usually again and again, you know, furnishing an inherent balance of exploration and exploitation like so so a lot of people point that out that they just find a lot more sort of credibility in the notion of like oh these agents because of gee uh they're sort of naturally internally you know architecturally incentivized to seek out more information. And nonetheless, like once they're affirmed that they have enough information, they are over time more than likely going to become like, oh no,

I've seen that before. So that they kind of know not to go necessarily in that direction. However, it's over a policy horizon. So maybe that policy horizon is five steps in the future or in continuous state space. It could be, you know, a minute into the future. uh and what they expect to kind of see over along the way. Um you know, so so even if they think they've explored a space well, they might not necessarily know the full sequence of transitions that'll happen over time. So that it's it's very likely that they'll be able to depending on your setup continuously find ways of like getting information gain out of a situation, computing that like incorporating that into their decision-making. But once they become much more reaffirmed that they've explored enough Then they start to, you know, I'm I'm speaking from from experience.

Sorry. Oh, in Monterey Bay.

I'm not sure he's talking to us. No. Uh here you go. Um so,

uh yeah, I that kind of interrupted my sorry train of thought, but it's just to get to the point. It's like very often than not I I think that T-maze example and spending just like an extra not a long time but just like a minute or two.

Um because I know with robotics you're usually looking at continuous streams, right? I mean maybe there's a I mean go sorry. Go ahead. Yeah, I was I was just going to say yes. Um, what I would love to be able to do is take a six degree of freedom robot arm and output the six

joint variables

uh that are continuous, you know, and u
between some limits and um that would
all be given some sensory input and then
beliefs about what that sensory input
indicates about current state.

Yeah. No, it's incredibly relevant to a
lot of the work that is done here. And
it's sort of, you know, one way that
they break down if you if you've looked
at the book and and become more familiar
with the notion of like an observation
modality, which is, you know, a very
common sensical thing. If you get rid of
the jargon, it's just sort of like, you
know, I can see things versus feel
things versus taste things. like there
there are different ways that I can
receive information and and so they tend
to break those up into three different
categories. And so there's um
exterceptive which would be like
information I receive from the outside
which includes all the things they said
earlier that the sort of common five
senses of like I can see and touch and
feel things. Interosceptive would be
like internal bodily signals um that are
coming like sort of from within that I'm
not necessarily feeling them in the in
the typical sense of like my skin is in
you know sort of dermal aspects. Um and
then there's proprioceptive and that
that's if you come across that term or
sorry by the way I don't mean to sound
condescending if you already are
familiar with it. No no this is great.
Thank you. Cool. Definitely. Yeah. So,

so proprioceptive I'm just trying to
find a quick and we might have to wrap
up then is
uh I know it's in
here. Um somewhere there's there's they
have a nice sort of visual description
of like the idea of like this
proprioceptive aforin. So this is you
know this is still related to more like
the brain but
proprioception you know it has much more
to do with like movement and kind of
reactive systems. So so there could be
like a reflex arc sort of logic for
example
um that that might be of interest to you
if you want a model that's kind of a
little bit more complex or sort of sort
of savvy with with putting together the
architecture in regards to active
inference. There's no necessity of it
though and I think you'll you'll you
might especially find chapter eight
interesting which is going to explicitly
go over continuous models. It of course
is different because of the continuous
space like time and continuous state space
sort of setup. Um but then in that case
we're looking at more like kind of these
attracting points or or here this is uh
this is rather complex but this is a
Lawrence you know chaotic system and
sort of like different kinds of dynamics
and viewing them in different ways and
viewing how you know like I mentioned
with the discrete state space setting
there's C which is sort of defining your
goals or your expected observations you

want um the the sort of equivalent of
that in continuous state space models.
You could view that as uh is sort of
like an attracting point or or a
distribution of attracting points or an
attracting distribution uh over a
continuous space than just modeling it
like that. Um you'll also it it relies a
lot more on sort of tailaylor series
approximations. There's this notion he
calls refers to it as the uh uh
generalized coordinates of motion. So
it's sort of like there these more sort
of working more with more directly and
explicitly with derivatives um over
time. So um yeah it's yeah I realized
that that actually kind of shot by the
time um but uh thanks for this uh
opportunity to get into uh to sort of
the deeper theoretics and and and really
honing in on some of these equations. Uh
yeah, you know, some Yeah, it's
sometimes folks with the here I'll I'll
uh pause. Um also, thanks for bearing
with the uh this is my first time ever
when I'm doing this purely solo,
so yeah. Um, so I'll go ahead and stop
recording and thank you everyone and see
you at the next See you at the next